

ASSARA System Architecture, Calculation Engines & Algorithmic Specifications

Complete technical treatise on geospatial calculations, biophysical atmospheric engines, soil nutrient balancing, phenological yield intervals, deterministic financial ROI, and multi-model AI decision routing.

Document Intent & Scope: This document provides an exhaustive, mathematically rigorous specification of the **ASSARA** decision-support platform. It details every mathematical formula, deterministic rule engine, geospatial calculation, and data integration workflow that transforms raw farmer input, satellite telemetry, and market prices into verified, explainable agricultural recommendations.

1. Core Product Loop & Architecture

ASSARA is built around an uncompromising principle: **Zero fake data, zero mock numbers, and zero ungrounded recommendations**. Every decision presented to the farmer is derived through a 9-step verified pipeline:

Step	Phase	Engine / Service	Input Data	Deterministic Output
1	USER	UserStore & Auth	Farmer profile, mobile/email, lang	Authenticated session & verified identity
2	FARM	FarmContext & Store	Farm plot name, registered plots	Active farm scope & field portfolio
3	LOCATION	LocationService	Browser GPS / Geocoding search	WGS-84 Lat/Lon, District, State, Centroid
4	BOUNDARY	Geospatial Engine	Drawn polygon vertices [lat, lon]	Geodesic Area (m², Hectares, Acres)
5	CROP	CropRegistry & Phenology	Crop, variety, sowing date, irrigation	Crop age (DAS), Phenological growth stage
6	TELEMETRY	Weather & Market Services	Open-Meteo GPS & Agmarknet APMC	Real-time Temp, Rain Prob, Mandi rates
7	RULES	Agriculture Rules Engine	Telemetry + Crop stage + Soil values	Thermal stress, Spray window, Irrigation, NPK
8	AI ADVISORY	AI Multi-Model Router	Verified Farm Context + Question	Structured [ACTION, WHY, WHEN, CONFIDENCE]
9	ACTION & DIARY	Intervention Journal	Executed spray/irrigation action	Persistent field diary & historical proof

2. Geospatial Calculation Engine

To ensure precise field measurements without planar distortion, ASSARA implements the **Spherical Shoelace Algorithm** over the WGS-84 terrestrial reference sphere (mean Earth radius $R = 6,378,137$ meters).

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2.1 Spherical Shoelace Formula for Polygon Area:
Given polygon vertices P = [(lat_0, lon_0), (lat_1, lon_1), ..., (lat_n-1, lon_n-1)] in radians:
Area (m2) = (R2 / 2) * | sum [ (lon_{i+1} - lon_{i-1}) * sin(lat_i) ] |
Area (Hectares) = Area (m2) / 10,000
Area (Acres) = Area (m2) / 4,046.8564224

2.2 Haversine Geodesic Distance Formula:
Given two coordinates (lat_1, lon_1) and (lat_2, lon_2):
d_lat = lat_2 - lat_1, d_lon = lon_2 - lon_1
a = sin2(d_lat/2) + cos(lat_1) * cos(lat_2) * sin2(d_lon/2)
distance_km = 2 * R * atan2(sqrt(a), sqrt(1-a))

2.3 Polygon Centroid Coordinates:
lat_center = (sum lat_i) / n, lon_center = (sum lon_i) / n
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3. Atmospheric & Biophysical Weather Risk Engine

Open-Meteo high-resolution hourly telemetry is evaluated through deterministic biophysical threshold models:

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3.1 Nocturnal Thermal Stress Degree-Hours (HSDH):
Plants undergo excessive dark respiration when nighttime temperatures exceed 25.0 deg C.
HSDH = sum_{t=20:00}^{06:00} max(0, T_night(t) - 25.0 deg C)
- T_night < 25 deg C: NO STRESS (Optimal carbohydrate translocation to grain/pods).
- 25 deg C <= T_night < 27 deg C: MODERATE STRESS (12% to 20% carbohydrate burn; flower thinning).
- T_night >= 27 deg C: SEVERE STRESS (28% to 42% flower abortion rate and reduced test grain weight).

3.2 Rain Wash-Off Risk Score (RWRS):
Evaluates probability and depth within the 4-to-6 hour chemical rainfast uptake window:
RWRS = min(100, RainProbability (%) + (Precipitation > 0 ? 50 : 0))
- RWRS < 20%: LOW RISK (Optimal rainfast absorption window).
- 20% <= RWRS < 40%: MODERATE RISK (Requires non-ionic surfactant/sticker).
- RWRS >= 40%: HIGH RISK (Do NOT spray; chemical wash-off and runoff hazard).

3.3 Wind Drift & Inversion Hazard:
- Wind > 15 km/h: SEVERE DRIFT (Off-target displacement).
- Wind < 2.5 km/h with T > 30 deg C: THERMAL INVERSION RISK (Trapped droplet vaporization).
- 5 km/h <= Wind <= 12 km/h: OPTIMAL LAMINAR CANOPY PENETRATION.
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4. Spray Window Suitability & Chemical Dosage Engine

A composite multi-parameter suitability score (0 to 100) dictates the daily spraying verdict:

Parameter	Optimal Range	Marginal Range	Unsuitable Threshold	Penalty Deduction
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Wind Speed	4 – 12 km/h	12 – 15 km/h	> 15 km/h or < 2 km/h	-40 points if >15 km/h
Rain Probability	< 20%	20 – 35%	> 35%	-50 points if >35%
Air Temperature	18 – 29 deg C	30 – 34 deg C	> 34 deg C	-40 points if >34 deg C
Relative Humidity	50 – 80%	35 – 50%	< 35%	-25 points if <35%

Dosage Math: For active farm area A (Acres):

- **Total Chemical Required (Liters):** $V_{chem} = (Dosage_{ml_per_acre} * A) / 1,000$ (e.g. 250 ml/ac * 5.2 ac = 1.30 Litres)
- **Total Spray Water Required (Liters):** $V_{water} = 150\text{ Liters/acre} * A$ (e.g. 150 L/ac * 5.2 ac = 780 Litres)

5. Soil Nutrient Balance & Fertilizer Optimization Engine

Translates target elemental Nitrogen (N), Phosphate (P2O5), and Potash (K2O) requirements into commercial 50kg bags of DAP, Neem-coated Urea, and MOP:

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5.1 Elemental Target Adjustments Based on Soil Test:
Target_N = Base_N * (Soil_N > 280 ? 0.8 : (Soil_N < 180 ? 1.25 : 1.0))
Target_P = Base_P * (Soil_P > 25 ? 0.75 : (Soil_P < 12 ? 1.30 : 1.0))
Target_K = Base_K * (Soil_K > 320 ? 0.80 : (Soil_K < 150 ? 1.25 : 1.0))

5.2 Commercial Product Formulation Decomposition:
- DAP (18-46-0) Bags (50kg): Supplies 23 kg P2O5 + 9 kg N per bag:
Bags_DAP = (Target_P * Acres) / 23.0
- Neem-Coated Urea (46% N) Bags (50kg): Supplies 23 kg N per bag:
Remaining_N = (Target_N * Acres) - (9.0 * Bags_DAP)
Bags_Urea = max(0, Remaining_N / 23.0)
- MOP (60% K2O) Bags (50kg): Supplies 30 kg K2O per bag:
Bags_MOP = (Target_K * Acres) / 30.0
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6. Irrigation Deficit & Pump Runtime Engine

Calculates root-zone moisture depletion and translates water volume into exact tube-well operating hours:

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6.1 Soil Moisture Deficit & Net Irrigation Depth:
Deficit (%) = max(0, Optimal_Moisture (70%) - Current_Moisture (%))
Depth_Required (mm) = (Deficit / 100) * 45.0 mm

6.2 Volume & Tube-Well Operating Runtime:
- Field Water Volume (Liters):  $V_{total} = (Depth_{mm} * 4,047\text{ L/ac*mm} * Acres) / Efficiency$ 
(Efficiency: Drip = 0.90, Sprinkler = 0.75, Flood = 0.50)
- Pump Operating Hours:  $Hours = V_{total} / (Pump_{HP} * 6,000\text{ Liters/Hour})$ 
(For a standard 5 HP agricultural submersible pump discharge = 30,000 L/hr)
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7. Deterministic Economics & ROBI Proof

All financial metrics are calculated via deterministic code with complete algebraic transparency:

Equation Step	Mathematical Formula	Example (Soybean 5.2ac @ Rs 4,850/q)
Gross Revenue	Gross = Yield (q/ac) * Mandi Price (Rs/q) * Acres	9.5 q/ac * Rs 4,850/q * 5.2 ac = Rs 2,39,590
Total Production Cost	Cost = sum(Seed + Fert + Protect + Labor + Irrig) * Acres	Rs 14,000/ac * 5.2 ac = Rs 72,800
Net Farm Profit	Net Profit = Gross Revenue - Total Production Cost	Rs 2,39,590 - Rs 72,800 = Rs 1,66,790
ROI Percentage	ROI (%) = (Net Farm Profit / Total Production Cost) * 100	(Rs 1,66,790 / Rs 72,800) * 100 = 229.1%
Protected Gain (ROBI)	ROBI Multiple = (d_Yield * Price * Acres) / (Cost_product + Labor)	(0.52 q/ac * Rs 4,850 * 5.2) / (Rs 570 * 5.2) = 4.42x

8. AI Multi-Model Router & Grounding Engine

ASSARA integrates Google Gemini 2.5 through an intelligent cost/latency router and rigid context injection layer:

- 8.1 Model Selection Hierarchy:**
 - **SIMPLE (Mandi price, Current temp, Dosage queries):** Gemini 2.5 Flash Lite (0.2s latency, low token cost).
 - **COMPLEX (Diagnosis, Multi-factor reasoning, What-If):** Gemini 2.5 Flash / Pro (Deep Agronomic CoT).
 - **IMAGE (Leaf pest & foliar disease photo diagnosis):** Gemini 2.5 Flash Multimodal Vision.
- 8.2 Grounded Context Injection Payload:**

Before prompt execution, the backend injects an immutable verified fact dictionary:
[Farmer, Plot Name, Measured Acres, Location Lat/Lon, Crop & DAS, Open-Meteo Weather, Agmarknet Mandi Price, Soil Test Status]
- 8.3 Enforced Decision Schema:**

Every recommendation MUST output:

 1. **ACTION:** Specific physical directive.
 2. **WHY:** Atmospheric and biological causal rationale.
 3. **WHEN:** Explicit time window (e.g. 4:30 PM - 6:45 PM).
 4. **CONFIDENCE:** High / Medium / Low.
 5. **DATA:** List of exact sensors and API sources used.
 6. **MISSING DATA GATE:** If information is insufficient, ask a clarifying question without fabricating.

ASSARA ARCHITECTURE VERIFICATION: All calculation engines, rules, APIs, and geospatial mathematics described in this document are fully implemented, compiled, and deployed in production.